

## Cumulative cutflows

Table 1: RunIII2024Summer24: expected event yields for the resolved, dimuon channel. Each row shows the yield after applying that cut and all preceding cuts. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                         | Z+jets             | $t\bar{t}$ +tW     | Nonprompt          | Other              | (2000,1100)        | (4000,2100)        | (6000,3100)        |
|---------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| No cuts                                           | $4.59 \times 10^8$ | $1.14 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $5.27 \times 10^4$ | $6.14 \times 10^2$ | $2.47 \times 10^1$ |
| $\geq 2$ AK4 jets ( $p_T > 40$ , $ \eta  < 2.4$ ) | $1.52 \times 10^8$ | $9.47 \times 10^6$ | $2.73 \times 10^8$ | $5.54 \times 10^6$ | $5.27 \times 10^4$ | $6.14 \times 10^2$ | $2.47 \times 10^1$ |
| $\geq 2$ AK4 jets (tight lepton veto)             | $9.10 \times 10^7$ | $9.36 \times 10^6$ | $2.54 \times 10^8$ | $5.41 \times 10^6$ | $5.25 \times 10^4$ | $6.12 \times 10^2$ | $2.46 \times 10^1$ |
| 2 muons ( $p_T > 53$ , $ \eta  < 2.4$ )           | $1.60 \times 10^6$ | $2.52 \times 10^5$ | $1.22 \times 10^6$ | $2.93 \times 10^4$ | $2.05 \times 10^4$ | $2.19 \times 10^2$ | 7.96               |
| 2 muons (tight ID)                                | $1.39 \times 10^6$ | $1.59 \times 10^5$ | $1.26 \times 10^4$ | $1.49 \times 10^4$ | $1.80 \times 10^4$ | $1.76 \times 10^2$ | 5.90               |
| Muon trigger                                      | $1.31 \times 10^6$ | $1.57 \times 10^5$ | $1.16 \times 10^4$ | $1.47 \times 10^4$ | $1.79 \times 10^4$ | $1.75 \times 10^2$ | 5.82               |
| $\Delta R > 0.4$ (all pairs)                      | $1.92 \times 10^5$ | $7.18 \times 10^4$ | $8.35 \times 10^2$ | $3.72 \times 10^3$ | $1.62 \times 10^4$ | $1.64 \times 10^2$ | 5.54               |
| Jet veto map                                      | $1.55 \times 10^5$ | $5.73 \times 10^4$ | $6.47 \times 10^2$ | $2.99 \times 10^3$ | $1.28 \times 10^4$ | $1.29 \times 10^2$ | 4.38               |
| $m(\ell\ell jj) > 800$ GeV                        | $3.00 \times 10^4$ | $7.37 \times 10^3$ | $1.25 \times 10^2$ | $5.39 \times 10^2$ | $1.28 \times 10^4$ | $1.29 \times 10^2$ | 4.38               |
| $m(\ell\ell) > 200$ GeV                           | $1.43 \times 10^3$ | $4.71 \times 10^3$ | $3.94 \times 10^1$ | $1.03 \times 10^2$ | $1.26 \times 10^4$ | $1.28 \times 10^2$ | 4.38               |
| $m(\ell\ell) > 400$ GeV                           | $4.99 \times 10^2$ | $1.54 \times 10^3$ | $1.04 \times 10^1$ | $5.08 \times 10^1$ | $1.16 \times 10^4$ | $1.25 \times 10^2$ | 4.35               |

Table 2: RunIII2024Summer24: expected event yields for the resolved, dielectron channel. Each row shows the yield after applying that cut and all preceding cuts. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                         | Z+jets             | $t\bar{t}$ +tW     | Nonprompt          | Other              | (2000,1100)        | (4000,2100)        | (6000,3100)        |
|---------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| No cuts                                           | $4.59 \times 10^8$ | $1.14 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $5.27 \times 10^4$ | $6.14 \times 10^2$ | $2.47 \times 10^1$ |
| $\geq 2$ AK4 jets ( $p_T > 40$ , $ \eta  < 2.4$ ) | $1.52 \times 10^8$ | $9.47 \times 10^6$ | $2.73 \times 10^8$ | $5.54 \times 10^6$ | $5.27 \times 10^4$ | $6.14 \times 10^2$ | $2.47 \times 10^1$ |
| $\geq 2$ AK4 jets (tight lepton veto)             | $9.10 \times 10^7$ | $9.36 \times 10^6$ | $2.54 \times 10^8$ | $5.41 \times 10^6$ | $5.25 \times 10^4$ | $6.12 \times 10^2$ | $2.46 \times 10^1$ |
| 2 electrons ( $p_T > 53$ , $ \eta  < 2.4$ )       | $1.03 \times 10^6$ | $2.40 \times 10^5$ | $1.20 \times 10^6$ | $3.03 \times 10^4$ | $1.87 \times 10^4$ | $2.20 \times 10^2$ | 8.93               |
| 2 electrons (HEEP ID)                             | $6.97 \times 10^5$ | $1.10 \times 10^5$ | $4.70 \times 10^3$ | $1.04 \times 10^4$ | $1.20 \times 10^4$ | $1.41 \times 10^2$ | 5.79               |
| Electron trigger                                  | $6.87 \times 10^5$ | $1.09 \times 10^5$ | $4.43 \times 10^3$ | $1.03 \times 10^4$ | $1.20 \times 10^4$ | $1.41 \times 10^2$ | 5.79               |
| $\Delta R > 0.4$ (all pairs)                      | $1.37 \times 10^5$ | $5.50 \times 10^4$ | $9.07 \times 10^2$ | $2.88 \times 10^3$ | $1.06 \times 10^4$ | $1.31 \times 10^2$ | 5.40               |
| Jet veto map                                      | $1.13 \times 10^5$ | $4.47 \times 10^4$ | $7.28 \times 10^2$ | $2.36 \times 10^3$ | $8.65 \times 10^3$ | $1.04 \times 10^2$ | 4.37               |
| $m(\ell\ell jj) > 800$ GeV                        | $1.96 \times 10^4$ | $5.13 \times 10^3$ | $1.62 \times 10^2$ | $4.04 \times 10^2$ | $8.65 \times 10^3$ | $1.04 \times 10^2$ | 4.37               |
| $m(\ell\ell) > 200$ GeV                           | $1.04 \times 10^3$ | $3.28 \times 10^3$ | $8.24 \times 10^1$ | $8.38 \times 10^1$ | $8.43 \times 10^3$ | $1.04 \times 10^2$ | 4.37               |
| $m(\ell\ell) > 400$ GeV                           | $3.51 \times 10^2$ | $1.05 \times 10^3$ | $3.21 \times 10^1$ | $3.79 \times 10^1$ | $7.69 \times 10^3$ | $1.03 \times 10^2$ | 4.35               |

Table 3: RunIII2024Summer24: expected event yields for the boosted, dimuon channel. Each row shows the yield after applying that cut and all preceding cuts. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                              | Z+jets             | $t\bar{t}$ +tW     | Nonprompt          | Other              | (2000,100)         | (4000,100)         | (6000,100)         |
|----------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| No cuts                                | $4.59 \times 10^8$ | $1.14 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.86 \times 10^4$ | $1.42 \times 10^3$ | $1.16 \times 10^2$ |
| Boosted tag                            | $4.59 \times 10^8$ | $1.11 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.65 \times 10^4$ | $1.40 \times 10^3$ | $1.13 \times 10^2$ |
| Leading lepton is $\mu$                | $2.21 \times 10^7$ | $2.10 \times 10^6$ | $6.22 \times 10^7$ | $5.28 \times 10^5$ | $4.40 \times 10^4$ | $6.36 \times 10^2$ | $4.90 \times 10^1$ |
| Leading tight lepton $p_T > 60$ GeV    | $1.29 \times 10^7$ | $1.73 \times 10^6$ | $4.07 \times 10^7$ | $4.02 \times 10^5$ | $4.39 \times 10^4$ | $6.33 \times 10^2$ | $4.86 \times 10^1$ |
| $\mu$ trigger fired                    | $1.21 \times 10^7$ | $1.64 \times 10^6$ | $3.82 \times 10^7$ | $3.78 \times 10^5$ | $4.30 \times 10^4$ | $6.16 \times 10^2$ | $4.72 \times 10^1$ |
| No $60 < m(\ell\ell) < 150$ GeV        | $1.14 \times 10^7$ | $1.62 \times 10^6$ | $3.81 \times 10^7$ | $3.70 \times 10^5$ | $4.30 \times 10^4$ | $6.15 \times 10^2$ | $4.71 \times 10^1$ |
| Merged AK8 jet with $\Delta\phi > 2.0$ | $8.06 \times 10^3$ | $2.52 \times 10^4$ | $5.64 \times 10^4$ | $1.14 \times 10^3$ | $2.84 \times 10^4$ | $3.96 \times 10^2$ | $2.58 \times 10^1$ |
| No extra tight lepton                  | $1.37 \times 10^3$ | $5.20 \times 10^3$ | $8.81 \times 10^2$ | $5.75 \times 10^1$ | $2.61 \times 10^4$ | $3.73 \times 10^2$ | $2.36 \times 10^1$ |
| Loose SF lepton in merged jet          | $1.37 \times 10^3$ | $5.20 \times 10^3$ | $8.81 \times 10^2$ | $5.75 \times 10^1$ | $2.61 \times 10^4$ | $3.73 \times 10^2$ | $2.36 \times 10^1$ |
| No loose OF lepton in merged jet       | $1.36 \times 10^3$ | $5.11 \times 10^3$ | $8.73 \times 10^2$ | $5.69 \times 10^1$ | $2.61 \times 10^4$ | $3.72 \times 10^2$ | $2.35 \times 10^1$ |
| $m(\ell\ell) > 200$ GeV                | $1.32 \times 10^3$ | $4.13 \times 10^3$ | $6.03 \times 10^2$ | $5.25 \times 10^1$ | $2.61 \times 10^4$ | $3.72 \times 10^2$ | $2.35 \times 10^1$ |
| $m(\ell J) > 800$ GeV                  | $1.12 \times 10^2$ | $5.70 \times 10^2$ | $7.35 \times 10^1$ | 7.72               | $2.59 \times 10^4$ | $3.63 \times 10^2$ | $2.20 \times 10^1$ |
| Jet veto map                           | $9.40 \times 10^1$ | $4.70 \times 10^2$ | $5.83 \times 10^1$ | 6.57               | $2.29 \times 10^4$ | $3.15 \times 10^2$ | $1.91 \times 10^1$ |

Table 4: RunIII2024Summer24: expected event yields for the boosted, dielectron channel. Each row shows the yield after applying that cut and all preceding cuts. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                              | Z+jets             | $t\bar{t}$ +tW     | Nonprompt          | Other              | (2000,100)         | (4000,100)         | (6000,100)         |
|----------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| No cuts                                | $4.59 \times 10^8$ | $1.14 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.86 \times 10^4$ | $1.42 \times 10^3$ | $1.16 \times 10^2$ |
| Boosted tag                            | $4.59 \times 10^8$ | $1.11 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.65 \times 10^4$ | $1.40 \times 10^3$ | $1.13 \times 10^2$ |
| Leading lepton is $e$                  | $1.85 \times 10^7$ | $1.79 \times 10^6$ | $5.21 \times 10^7$ | $4.48 \times 10^5$ | $4.05 \times 10^4$ | $5.62 \times 10^2$ | $4.42 \times 10^1$ |
| Leading tight lepton $p_T > 60$ GeV    | $1.09 \times 10^7$ | $1.48 \times 10^6$ | $3.43 \times 10^7$ | $3.42 \times 10^5$ | $4.04 \times 10^4$ | $5.60 \times 10^2$ | $4.36 \times 10^1$ |
| $e$ trigger fired                      | $1.00 \times 10^7$ | $1.36 \times 10^6$ | $3.06 \times 10^7$ | $3.11 \times 10^5$ | $4.03 \times 10^4$ | $5.58 \times 10^2$ | $4.34 \times 10^1$ |
| No $60 < m(\ell\ell) < 150$ GeV        | $9.46 \times 10^6$ | $1.34 \times 10^6$ | $3.06 \times 10^7$ | $3.05 \times 10^5$ | $4.03 \times 10^4$ | $5.58 \times 10^2$ | $4.33 \times 10^1$ |
| Merged AK8 jet with $\Delta\phi > 2.0$ | $7.47 \times 10^3$ | $2.16 \times 10^4$ | $4.80 \times 10^4$ | $1.03 \times 10^3$ | $2.33 \times 10^4$ | $2.33 \times 10^2$ | $1.62 \times 10^1$ |
| No extra tight lepton                  | $1.36 \times 10^3$ | $5.77 \times 10^3$ | $1.79 \times 10^3$ | $8.92 \times 10^1$ | $1.09 \times 10^4$ | $5.95 \times 10^1$ | 6.50               |
| Loose SF lepton in merged jet          | $1.36 \times 10^3$ | $5.77 \times 10^3$ | $1.79 \times 10^3$ | $8.92 \times 10^1$ | $1.09 \times 10^4$ | $5.95 \times 10^1$ | 6.50               |
| No loose OF lepton in merged jet       | $1.36 \times 10^3$ | $5.66 \times 10^3$ | $1.78 \times 10^3$ | $8.88 \times 10^1$ | $1.08 \times 10^4$ | $5.91 \times 10^1$ | 6.48               |
| $m(\ell\ell) > 200$ GeV                | $1.31 \times 10^3$ | $4.59 \times 10^3$ | $1.52 \times 10^3$ | $8.30 \times 10^1$ | $1.08 \times 10^4$ | $5.91 \times 10^1$ | 6.48               |
| $m(\ell J) > 800$ GeV                  | $1.22 \times 10^2$ | $5.00 \times 10^2$ | $1.52 \times 10^2$ | $1.64 \times 10^1$ | $1.06 \times 10^4$ | $5.26 \times 10^1$ | 5.33               |
| Jet veto map                           | $1.05 \times 10^2$ | $4.17 \times 10^2$ | $1.23 \times 10^2$ | $1.38 \times 10^1$ | $9.57 \times 10^3$ | $4.60 \times 10^1$ | 4.76               |

## One-cut cutflows

Table 5: RunIII2024Summer24: expected event yields for the resolved, dimuon channel. Each row shows the yield after applying only that single cut, independent of all other cuts. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                         | Z+jets             | $t\bar{t}$ +tW     | Nonprompt          | Other              | (2000,1100)        | (4000,2100)        | (6000,3100)        |
|---------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| No cuts                                           | $4.59 \times 10^8$ | $1.14 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $5.27 \times 10^4$ | $6.14 \times 10^2$ | $2.47 \times 10^1$ |
| $\geq 2$ AK4 jets ( $p_T > 40$ , $ \eta  < 2.4$ ) | $1.52 \times 10^8$ | $9.47 \times 10^6$ | $2.73 \times 10^8$ | $5.54 \times 10^6$ | $5.27 \times 10^4$ | $6.14 \times 10^2$ | $2.47 \times 10^1$ |
| $\geq 2$ AK4 jets (tight lepton veto)             | $2.92 \times 10^8$ | $1.12 \times 10^7$ | $1.24 \times 10^9$ | $1.26 \times 10^7$ | $5.25 \times 10^4$ | $6.12 \times 10^2$ | $2.46 \times 10^1$ |
| 2 muons ( $p_T > 53$ , $ \eta  < 2.4$ )           | $2.62 \times 10^6$ | $2.63 \times 10^5$ | $2.41 \times 10^6$ | $4.20 \times 10^4$ | $2.06 \times 10^4$ | $2.20 \times 10^2$ | 7.98               |
| 2 muons (tight ID)                                | $9.86 \times 10^7$ | $1.13 \times 10^6$ | $1.42 \times 10^6$ | $1.89 \times 10^5$ | $2.16 \times 10^4$ | $2.43 \times 10^2$ | 9.61               |
| Muon trigger                                      | $2.90 \times 10^7$ | $2.61 \times 10^6$ | $7.31 \times 10^7$ | $5.95 \times 10^5$ | $2.72 \times 10^4$ | $3.17 \times 10^2$ | $1.28 \times 10^1$ |
| $\Delta R > 0.4$ (all pairs)                      | $3.54 \times 10^5$ | $2.70 \times 10^5$ | $3.91 \times 10^3$ | $8.82 \times 10^3$ | $3.36 \times 10^4$ | $4.12 \times 10^2$ | $1.68 \times 10^1$ |
| Jet veto map                                      | $4.20 \times 10^8$ | $9.44 \times 10^6$ | $1.37 \times 10^9$ | $1.17 \times 10^7$ | $4.22 \times 10^4$ | $4.88 \times 10^2$ | $1.96 \times 10^1$ |
| $m(\ell\ell jj) > 800$ GeV                        | $7.82 \times 10^4$ | $4.14 \times 10^4$ | $2.07 \times 10^3$ | $2.04 \times 10^3$ | $3.47 \times 10^4$ | $4.25 \times 10^2$ | $1.74 \times 10^1$ |
| $m(\ell\ell) > 200$ GeV                           | $2.61 \times 10^5$ | $2.14 \times 10^5$ | $6.03 \times 10^3$ | $1.71 \times 10^4$ | $3.67 \times 10^4$ | $4.41 \times 10^2$ | $1.78 \times 10^1$ |
| $m(\ell\ell) > 400$ GeV                           | $2.58 \times 10^4$ | $2.26 \times 10^4$ | $7.85 \times 10^2$ | $2.90 \times 10^3$ | $3.34 \times 10^4$ | $4.31 \times 10^2$ | $1.76 \times 10^1$ |

Table 6: RunIII2024Summer24: expected event yields for the resolved, dielectron channel. Each row shows the yield after applying only that single cut, independent of all other cuts. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                         | Z+jets             | $t\bar{t}$ +tW     | Nonprompt          | Other              | (2000,1100)        | (4000,2100)        | (6000,3100)        |
|---------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| No cuts                                           | $4.59 \times 10^8$ | $1.14 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $5.27 \times 10^4$ | $6.14 \times 10^2$ | $2.47 \times 10^1$ |
| $\geq 2$ AK4 jets ( $p_T > 40$ , $ \eta  < 2.4$ ) | $1.52 \times 10^8$ | $9.47 \times 10^6$ | $2.73 \times 10^8$ | $5.54 \times 10^6$ | $5.27 \times 10^4$ | $6.14 \times 10^2$ | $2.47 \times 10^1$ |
| $\geq 2$ AK4 jets (tight lepton veto)             | $2.92 \times 10^8$ | $1.12 \times 10^7$ | $1.24 \times 10^9$ | $1.26 \times 10^7$ | $5.25 \times 10^4$ | $6.12 \times 10^2$ | $2.46 \times 10^1$ |
| 2 electrons ( $p_T > 53$ , $ \eta  < 2.4$ )       | $1.47 \times 10^6$ | $2.46 \times 10^5$ | $1.28 \times 10^6$ | $3.58 \times 10^4$ | $1.87 \times 10^4$ | $2.21 \times 10^2$ | 8.94               |
| 2 electrons (HEEP ID)                             | $3.51 \times 10^7$ | $2.88 \times 10^5$ | $2.46 \times 10^4$ | $5.35 \times 10^4$ | $1.57 \times 10^4$ | $1.90 \times 10^2$ | 7.79               |
| Electron trigger                                  | $1.07 \times 10^8$ | $3.43 \times 10^6$ | $1.63 \times 10^8$ | $9.95 \times 10^5$ | $2.92 \times 10^4$ | $3.65 \times 10^2$ | $1.49 \times 10^1$ |
| $\Delta R > 0.4$ (all pairs)                      | $3.54 \times 10^5$ | $2.70 \times 10^5$ | $3.91 \times 10^3$ | $8.82 \times 10^3$ | $3.36 \times 10^4$ | $4.12 \times 10^2$ | $1.68 \times 10^1$ |
| Jet veto map                                      | $4.20 \times 10^8$ | $9.44 \times 10^6$ | $1.37 \times 10^9$ | $1.17 \times 10^7$ | $4.22 \times 10^4$ | $4.88 \times 10^2$ | $1.96 \times 10^1$ |
| $m(\ell\ell jj) > 800$ GeV                        | $7.82 \times 10^4$ | $4.14 \times 10^4$ | $2.07 \times 10^3$ | $2.04 \times 10^3$ | $3.47 \times 10^4$ | $4.25 \times 10^2$ | $1.74 \times 10^1$ |
| $m(\ell\ell) > 200$ GeV                           | $2.61 \times 10^5$ | $2.14 \times 10^5$ | $6.03 \times 10^3$ | $1.71 \times 10^4$ | $3.67 \times 10^4$ | $4.41 \times 10^2$ | $1.78 \times 10^1$ |
| $m(\ell\ell) > 400$ GeV                           | $2.58 \times 10^4$ | $2.26 \times 10^4$ | $7.85 \times 10^2$ | $2.90 \times 10^3$ | $3.34 \times 10^4$ | $4.31 \times 10^2$ | $1.76 \times 10^1$ |

Table 7: RunIII2024Summer24: expected event yields for the boosted, dimuon channel. Each row shows the yield after applying only that single cut, independent of all other cuts. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                              | Z+jets             | $t\bar{t}$ +tW     | Nonprompt          | Other              | (2000,100)         | (4000,100)         | (6000,100)         |
|----------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| No cuts                                | $4.59 \times 10^8$ | $1.14 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.86 \times 10^4$ | $1.42 \times 10^3$ | $1.16 \times 10^2$ |
| Boosted tag                            | $4.59 \times 10^8$ | $1.11 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.65 \times 10^4$ | $1.40 \times 10^3$ | $1.13 \times 10^2$ |
| Leading lepton is $\mu$                | $2.23 \times 10^7$ | $2.24 \times 10^6$ | $6.22 \times 10^7$ | $5.33 \times 10^5$ | $4.54 \times 10^4$ | $6.48 \times 10^2$ | $5.10 \times 10^1$ |
| Leading tight lepton $p_T > 60$ GeV    | $2.41 \times 10^7$ | $3.47 \times 10^6$ | $7.50 \times 10^7$ | $7.53 \times 10^5$ | $8.63 \times 10^4$ | $1.21 \times 10^3$ | $9.54 \times 10^1$ |
| $\mu$ trigger fired                    | $2.90 \times 10^7$ | $2.61 \times 10^6$ | $7.31 \times 10^7$ | $5.95 \times 10^5$ | $4.66 \times 10^4$ | $6.64 \times 10^2$ | $5.25 \times 10^1$ |
| No $60 < m(\ell\ell) < 150$ GeV        | $3.90 \times 10^7$ | $4.06 \times 10^6$ | $1.14 \times 10^8$ | $9.63 \times 10^5$ | $8.65 \times 10^4$ | $1.21 \times 10^3$ | $9.62 \times 10^1$ |
| Merged AK8 jet with $\Delta\phi > 2.0$ | $2.28 \times 10^4$ | $8.01 \times 10^4$ | $1.19 \times 10^5$ | $2.73 \times 10^3$ | $5.34 \times 10^4$ | $6.48 \times 10^2$ | $4.43 \times 10^1$ |
| No extra tight lepton                  | $3.71 \times 10^3$ | $2.56 \times 10^4$ | $3.22 \times 10^3$ | $2.14 \times 10^2$ | $3.85 \times 10^4$ | $4.49 \times 10^2$ | $3.22 \times 10^1$ |
| Loose SF lepton in merged jet          | $3.79 \times 10^3$ | $2.59 \times 10^4$ | $3.23 \times 10^3$ | $2.92 \times 10^2$ | $3.85 \times 10^4$ | $4.49 \times 10^2$ | $3.23 \times 10^1$ |
| No loose OF lepton in merged jet       | $4.59 \times 10^8$ | $1.13 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.83 \times 10^4$ | $1.42 \times 10^3$ | $1.16 \times 10^2$ |
| $m(\ell\ell) > 200$ GeV                | $3.61 \times 10^3$ | $3.74 \times 10^4$ | $4.85 \times 10^3$ | $4.86 \times 10^2$ | $3.86 \times 10^4$ | $4.53 \times 10^2$ | $3.24 \times 10^1$ |
| $m(\ell J) > 800$ GeV                  | $2.23 \times 10^3$ | $5.50 \times 10^3$ | $1.22 \times 10^4$ | $4.58 \times 10^2$ | $5.29 \times 10^4$ | $6.25 \times 10^2$ | $4.01 \times 10^1$ |
| Jet veto map                           | $4.20 \times 10^8$ | $9.44 \times 10^6$ | $1.37 \times 10^9$ | $1.17 \times 10^7$ | $8.67 \times 10^4$ | $1.24 \times 10^3$ | $1.02 \times 10^2$ |

Table 8: RunIII2024Summer24: expected event yields for the boosted, dielectron channel. Each row shows the yield after applying only that single cut, independent of all other cuts. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                              | Z+jets             | $t\bar{t}$ +tW     | Nonprompt          | Other              | (2000,100)         | (4000,100)         | (6000,100)         |
|----------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| No cuts                                | $4.59 \times 10^8$ | $1.14 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.86 \times 10^4$ | $1.42 \times 10^3$ | $1.16 \times 10^2$ |
| Boosted tag                            | $4.59 \times 10^8$ | $1.11 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.65 \times 10^4$ | $1.40 \times 10^3$ | $1.13 \times 10^2$ |
| Leading lepton is $e$                  | $1.87 \times 10^7$ | $1.91 \times 10^6$ | $5.21 \times 10^7$ | $4.51 \times 10^5$ | $4.11 \times 10^4$ | $5.69 \times 10^2$ | $4.53 \times 10^1$ |
| Leading tight lepton $p_T > 60$ GeV    | $2.41 \times 10^7$ | $3.47 \times 10^6$ | $7.50 \times 10^7$ | $7.53 \times 10^5$ | $8.63 \times 10^4$ | $1.21 \times 10^3$ | $9.54 \times 10^1$ |
| $e$ trigger fired                      | $1.07 \times 10^8$ | $3.43 \times 10^6$ | $1.63 \times 10^8$ | $9.95 \times 10^5$ | $5.19 \times 10^4$ | $7.37 \times 10^2$ | $5.79 \times 10^1$ |
| No $60 < m(\ell\ell) < 150$ GeV        | $3.90 \times 10^7$ | $4.06 \times 10^6$ | $1.14 \times 10^8$ | $9.63 \times 10^5$ | $8.65 \times 10^4$ | $1.21 \times 10^3$ | $9.62 \times 10^1$ |
| Merged AK8 jet with $\Delta\phi > 2.0$ | $2.28 \times 10^4$ | $8.01 \times 10^4$ | $1.19 \times 10^5$ | $2.73 \times 10^3$ | $5.34 \times 10^4$ | $6.48 \times 10^2$ | $4.43 \times 10^1$ |
| No extra tight lepton                  | $3.71 \times 10^3$ | $2.56 \times 10^4$ | $3.22 \times 10^3$ | $2.14 \times 10^2$ | $3.85 \times 10^4$ | $4.49 \times 10^2$ | $3.22 \times 10^1$ |
| Loose SF lepton in merged jet          | $3.79 \times 10^3$ | $2.59 \times 10^4$ | $3.23 \times 10^3$ | $2.92 \times 10^2$ | $3.85 \times 10^4$ | $4.49 \times 10^2$ | $3.23 \times 10^1$ |
| No loose OF lepton in merged jet       | $4.59 \times 10^8$ | $1.13 \times 10^7$ | $1.50 \times 10^9$ | $1.33 \times 10^7$ | $9.83 \times 10^4$ | $1.42 \times 10^3$ | $1.16 \times 10^2$ |
| $m(\ell\ell) > 200$ GeV                | $3.61 \times 10^3$ | $3.74 \times 10^4$ | $4.85 \times 10^3$ | $4.86 \times 10^2$ | $3.86 \times 10^4$ | $4.53 \times 10^2$ | $3.24 \times 10^1$ |
| $m(\ell J) > 800$ GeV                  | $2.23 \times 10^3$ | $5.50 \times 10^3$ | $1.22 \times 10^4$ | $4.58 \times 10^2$ | $5.29 \times 10^4$ | $6.25 \times 10^2$ | $4.01 \times 10^1$ |
| Jet veto map                           | $4.20 \times 10^8$ | $9.44 \times 10^6$ | $1.37 \times 10^9$ | $1.17 \times 10^7$ | $8.67 \times 10^4$ | $1.24 \times 10^3$ | $1.02 \times 10^2$ |

## Running efficiency

Table 9: RunIII2024Summer24: expected event yields for the resolved, dimuon channel. Each cell shows the running efficiency (%) relative to the first selection row. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                      | Z+jets                | $t\bar{t}$ +tW        | Nonprompt             | Other                 | (2000,1100)         | (4000,2100)         | (6000,3100)         |
|------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|---------------------|---------------------|---------------------|
| No cuts                                        | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$  | $1.00 \times 10^2$  | $1.00 \times 10^2$  |
| $\geq 2$ AK4 jets ( $p_T > 40,  \eta  < 2.4$ ) | $3.31 \times 10^1$    | $8.34 \times 10^1$    | $1.82 \times 10^1$    | $4.15 \times 10^1$    | $10.00 \times 10^1$ | $10.00 \times 10^1$ | $10.00 \times 10^1$ |
| $\geq 2$ AK4 jets (tight lepton veto)          | $1.98 \times 10^1$    | $8.25 \times 10^1$    | $1.69 \times 10^1$    | $4.05 \times 10^1$    | $9.96 \times 10^1$  | $9.98 \times 10^1$  | $9.98 \times 10^1$  |
| 2 muons ( $p_T > 53,  \eta  < 2.4$ )           | $3.47 \times 10^{-1}$ | $2.22 \times 10^0$    | $8.12 \times 10^{-2}$ | $2.19 \times 10^{-1}$ | $3.89 \times 10^1$  | $3.57 \times 10^1$  | $3.23 \times 10^1$  |
| 2 muons (tight ID)                             | $3.04 \times 10^{-1}$ | $1.40 \times 10^0$    | $8.39 \times 10^{-4}$ | $1.12 \times 10^{-1}$ | $3.41 \times 10^1$  | $2.87 \times 10^1$  | $2.39 \times 10^1$  |
| Muon trigger                                   | $2.84 \times 10^{-1}$ | $1.38 \times 10^0$    | $7.71 \times 10^{-4}$ | $1.10 \times 10^{-1}$ | $3.40 \times 10^1$  | $2.86 \times 10^1$  | $2.36 \times 10^1$  |
| $\Delta R > 0.4$ (all pairs)                   | $4.17 \times 10^{-2}$ | $6.33 \times 10^{-1}$ | $5.55 \times 10^{-5}$ | $2.79 \times 10^{-2}$ | $3.07 \times 10^1$  | $2.67 \times 10^1$  | $2.25 \times 10^1$  |
| Jet veto map                                   | $3.38 \times 10^{-2}$ | $5.05 \times 10^{-1}$ | $4.30 \times 10^{-5}$ | $2.24 \times 10^{-2}$ | $2.43 \times 10^1$  | $2.09 \times 10^1$  | $1.78 \times 10^1$  |
| $m(\ell\ell jj) > 800$ GeV                     | $6.53 \times 10^{-3}$ | $6.49 \times 10^{-2}$ | $8.33 \times 10^{-6}$ | $4.04 \times 10^{-3}$ | $2.43 \times 10^1$  | $2.09 \times 10^1$  | $1.78 \times 10^1$  |
| $m(\ell\ell) > 200$ GeV                        | $3.11 \times 10^{-4}$ | $4.15 \times 10^{-2}$ | $2.62 \times 10^{-6}$ | $7.72 \times 10^{-4}$ | $2.39 \times 10^1$  | $2.09 \times 10^1$  | $1.77 \times 10^1$  |
| $m(\ell\ell) > 400$ GeV                        | $1.09 \times 10^{-4}$ | $1.35 \times 10^{-2}$ | $6.89 \times 10^{-7}$ | $3.81 \times 10^{-4}$ | $2.19 \times 10^1$  | $2.04 \times 10^1$  | $1.76 \times 10^1$  |

Table 10: RunIII2024Summer24: expected event yields for the resolved, dielectron channel. Each cell shows the running efficiency (%) relative to the first selection row. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                      | Z+jets                | $t\bar{t}$ +tW        | Nonprompt             | Other                 | (2000,1100)         | (4000,2100)         | (6000,3100)         |
|------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|---------------------|---------------------|---------------------|
| No cuts                                        | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$  | $1.00 \times 10^2$  | $1.00 \times 10^2$  |
| $\geq 2$ AK4 jets ( $p_T > 40,  \eta  < 2.4$ ) | $3.31 \times 10^1$    | $8.34 \times 10^1$    | $1.82 \times 10^1$    | $4.15 \times 10^1$    | $10.00 \times 10^1$ | $10.00 \times 10^1$ | $10.00 \times 10^1$ |
| $\geq 2$ AK4 jets (tight lepton veto)          | $1.98 \times 10^1$    | $8.25 \times 10^1$    | $1.69 \times 10^1$    | $4.05 \times 10^1$    | $9.96 \times 10^1$  | $9.98 \times 10^1$  | $9.98 \times 10^1$  |
| 2 electrons ( $p_T > 53,  \eta  < 2.4$ )       | $2.25 \times 10^{-1}$ | $2.12 \times 10^0$    | $7.96 \times 10^{-2}$ | $2.27 \times 10^{-1}$ | $3.54 \times 10^1$  | $3.59 \times 10^1$  | $3.62 \times 10^1$  |
| 2 electrons (HEEP ID)                          | $1.52 \times 10^{-1}$ | $9.69 \times 10^{-1}$ | $3.13 \times 10^{-4}$ | $7.79 \times 10^{-2}$ | $2.27 \times 10^1$  | $2.30 \times 10^1$  | $2.35 \times 10^1$  |
| Electron trigger                               | $1.50 \times 10^{-1}$ | $9.59 \times 10^{-1}$ | $2.95 \times 10^{-4}$ | $7.71 \times 10^{-2}$ | $2.27 \times 10^1$  | $2.30 \times 10^1$  | $2.35 \times 10^1$  |
| $\Delta R > 0.4$ (all pairs)                   | $2.97 \times 10^{-2}$ | $4.84 \times 10^{-1}$ | $6.03 \times 10^{-5}$ | $2.16 \times 10^{-2}$ | $2.01 \times 10^1$  | $2.13 \times 10^1$  | $2.19 \times 10^1$  |
| Jet veto map                                   | $2.46 \times 10^{-2}$ | $3.93 \times 10^{-1}$ | $4.84 \times 10^{-5}$ | $1.77 \times 10^{-2}$ | $1.64 \times 10^1$  | $1.70 \times 10^1$  | $1.77 \times 10^1$  |
| $m(\ell\ell jj) > 800$ GeV                     | $4.28 \times 10^{-3}$ | $4.52 \times 10^{-2}$ | $1.08 \times 10^{-5}$ | $3.02 \times 10^{-3}$ | $1.64 \times 10^1$  | $1.70 \times 10^1$  | $1.77 \times 10^1$  |
| $m(\ell\ell) > 200$ GeV                        | $2.27 \times 10^{-4}$ | $2.89 \times 10^{-2}$ | $5.48 \times 10^{-6}$ | $6.28 \times 10^{-4}$ | $1.60 \times 10^1$  | $1.70 \times 10^1$  | $1.77 \times 10^1$  |
| $m(\ell\ell) > 400$ GeV                        | $7.64 \times 10^{-5}$ | $9.25 \times 10^{-3}$ | $2.13 \times 10^{-6}$ | $2.84 \times 10^{-4}$ | $1.46 \times 10^1$  | $1.67 \times 10^1$  | $1.76 \times 10^1$  |

Table 11: RunIII2024Summer24: expected event yields for the boosted, dimuon channel. Each cell shows the running efficiency (%) relative to the first selection row. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                              | Z+jets                | $t\bar{t}$ +tW        | Nonprompt             | Other                 | (2000,100)         | (4000,100)         | (6000,100)         |
|----------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|--------------------|--------------------|--------------------|
| No cuts                                | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$ | $1.00 \times 10^2$ | $1.00 \times 10^2$ |
| Boosted tag                            | $9.99 \times 10^1$    | $9.76 \times 10^1$    | $10.00 \times 10^1$   | $9.99 \times 10^1$    | $9.79 \times 10^1$ | $9.87 \times 10^1$ | $9.73 \times 10^1$ |
| Leading lepton is $\mu$                | $4.81 \times 10^0$    | $1.85 \times 10^1$    | $4.13 \times 10^0$    | $3.96 \times 10^0$    | $4.46 \times 10^1$ | $4.47 \times 10^1$ | $4.21 \times 10^1$ |
| Leading tight lepton $p_T > 60$ GeV    | $2.80 \times 10^0$    | $1.52 \times 10^1$    | $2.71 \times 10^0$    | $3.01 \times 10^0$    | $4.45 \times 10^1$ | $4.45 \times 10^1$ | $4.17 \times 10^1$ |
| $\mu$ trigger fired                    | $2.64 \times 10^0$    | $1.45 \times 10^1$    | $2.54 \times 10^0$    | $2.83 \times 10^0$    | $4.36 \times 10^1$ | $4.33 \times 10^1$ | $4.05 \times 10^1$ |
| No $60 < m(\ell\ell) < 150$ GeV        | $2.48 \times 10^0$    | $1.43 \times 10^1$    | $2.54 \times 10^0$    | $2.78 \times 10^0$    | $4.36 \times 10^1$ | $4.32 \times 10^1$ | $4.04 \times 10^1$ |
| Merged AK8 jet with $\Delta\phi > 2.0$ | $1.75 \times 10^{-3}$ | $2.22 \times 10^{-1}$ | $3.75 \times 10^{-3}$ | $8.55 \times 10^{-3}$ | $2.88 \times 10^1$ | $2.78 \times 10^1$ | $2.21 \times 10^1$ |
| No extra tight lepton                  | $2.98 \times 10^{-4}$ | $4.58 \times 10^{-2}$ | $5.86 \times 10^{-5}$ | $4.31 \times 10^{-4}$ | $2.65 \times 10^1$ | $2.62 \times 10^1$ | $2.02 \times 10^1$ |
| Loose SF lepton in merged jet          | $2.98 \times 10^{-4}$ | $4.58 \times 10^{-2}$ | $5.86 \times 10^{-5}$ | $4.31 \times 10^{-4}$ | $2.65 \times 10^1$ | $2.62 \times 10^1$ | $2.02 \times 10^1$ |
| No loose OF lepton in merged jet       | $2.97 \times 10^{-4}$ | $4.50 \times 10^{-2}$ | $5.80 \times 10^{-5}$ | $4.27 \times 10^{-4}$ | $2.65 \times 10^1$ | $2.62 \times 10^1$ | $2.02 \times 10^1$ |
| $m(\ell\ell) > 200$ GeV                | $2.87 \times 10^{-4}$ | $3.64 \times 10^{-2}$ | $4.01 \times 10^{-5}$ | $3.94 \times 10^{-4}$ | $2.65 \times 10^1$ | $2.62 \times 10^1$ | $2.02 \times 10^1$ |
| $m(\ell J) > 800$ GeV                  | $2.44 \times 10^{-5}$ | $5.03 \times 10^{-3}$ | $4.89 \times 10^{-6}$ | $5.78 \times 10^{-5}$ | $2.63 \times 10^1$ | $2.55 \times 10^1$ | $1.89 \times 10^1$ |
| Jet veto map                           | $2.05 \times 10^{-5}$ | $4.14 \times 10^{-3}$ | $3.88 \times 10^{-6}$ | $4.92 \times 10^{-5}$ | $2.32 \times 10^1$ | $2.21 \times 10^1$ | $1.64 \times 10^1$ |

Table 12: RunIII2024Summer24: expected event yields for the boosted, dielectron channel. Each cell shows the running efficiency (%) relative to the first selection row. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                              | Z+jets                | $t\bar{t}$ +tW        | Nonprompt             | Other                 | (2000,100)         | (4000,100)         | (6000,100)         |
|----------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|--------------------|--------------------|--------------------|
| No cuts                                | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$ | $1.00 \times 10^2$ | $1.00 \times 10^2$ |
| Boosted tag                            | $9.99 \times 10^1$    | $9.76 \times 10^1$    | $10.00 \times 10^1$   | $9.99 \times 10^1$    | $9.79 \times 10^1$ | $9.87 \times 10^1$ | $9.73 \times 10^1$ |
| Leading lepton is $e$                  | $4.03 \times 10^0$    | $1.57 \times 10^1$    | $3.46 \times 10^0$    | $3.35 \times 10^0$    | $4.11 \times 10^1$ | $3.95 \times 10^1$ | $3.79 \times 10^1$ |
| Leading tight lepton $p_T > 60$ GeV    | $2.37 \times 10^0$    | $1.30 \times 10^1$    | $2.28 \times 10^0$    | $2.56 \times 10^0$    | $4.09 \times 10^1$ | $3.94 \times 10^1$ | $3.74 \times 10^1$ |
| $e$ trigger fired                      | $2.18 \times 10^0$    | $1.20 \times 10^1$    | $2.03 \times 10^0$    | $2.33 \times 10^0$    | $4.09 \times 10^1$ | $3.93 \times 10^1$ | $3.72 \times 10^1$ |
| No $60 < m(\ell\ell) < 150$ GeV        | $2.06 \times 10^0$    | $1.18 \times 10^1$    | $2.03 \times 10^0$    | $2.28 \times 10^0$    | $4.08 \times 10^1$ | $3.92 \times 10^1$ | $3.72 \times 10^1$ |
| Merged AK8 jet with $\Delta\phi > 2.0$ | $1.63 \times 10^{-3}$ | $1.90 \times 10^{-1}$ | $3.19 \times 10^{-3}$ | $7.70 \times 10^{-3}$ | $2.37 \times 10^1$ | $1.64 \times 10^1$ | $1.39 \times 10^1$ |
| No extra tight lepton                  | $2.95 \times 10^{-4}$ | $5.09 \times 10^{-2}$ | $1.19 \times 10^{-4}$ | $6.68 \times 10^{-4}$ | $1.10 \times 10^1$ | $4.18 \times 10^0$ | $5.58 \times 10^0$ |
| Loose SF lepton in merged jet          | $2.95 \times 10^{-4}$ | $5.09 \times 10^{-2}$ | $1.19 \times 10^{-4}$ | $6.68 \times 10^{-4}$ | $1.10 \times 10^1$ | $4.18 \times 10^0$ | $5.58 \times 10^0$ |
| No loose OF lepton in merged jet       | $2.95 \times 10^{-4}$ | $4.99 \times 10^{-2}$ | $1.18 \times 10^{-4}$ | $6.65 \times 10^{-4}$ | $1.10 \times 10^1$ | $4.16 \times 10^0$ | $5.56 \times 10^0$ |
| $m(\ell\ell) > 200$ GeV                | $2.85 \times 10^{-4}$ | $4.04 \times 10^{-2}$ | $1.01 \times 10^{-4}$ | $6.22 \times 10^{-4}$ | $1.10 \times 10^1$ | $4.16 \times 10^0$ | $5.56 \times 10^0$ |
| $m(\ell J) > 800$ GeV                  | $2.66 \times 10^{-5}$ | $4.40 \times 10^{-3}$ | $1.01 \times 10^{-5}$ | $1.23 \times 10^{-4}$ | $1.08 \times 10^1$ | $3.70 \times 10^0$ | $4.57 \times 10^0$ |
| Jet veto map                           | $2.29 \times 10^{-5}$ | $3.67 \times 10^{-3}$ | $8.17 \times 10^{-6}$ | $1.04 \times 10^{-4}$ | $9.71 \times 10^0$ | $3.23 \times 10^0$ | $4.09 \times 10^0$ |



## Relative efficiency

Table 13: RunIII2024Summer24: expected event yields for the resolved, dimuon channel. Each cell shows the relative efficiency (%): the fraction of events passing the previous cut that also pass the current cut. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                      | Z+jets             | $t\bar{t}$ +tW     | Nonprompt             | Other                 | (2000,1100)         | (4000,2100)         | (6000,3100)         |
|------------------------------------------------|--------------------|--------------------|-----------------------|-----------------------|---------------------|---------------------|---------------------|
| No cuts                                        | $1.00 \times 10^2$ | $1.00 \times 10^2$ | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$  | $1.00 \times 10^2$  | $1.00 \times 10^2$  |
| $\geq 2$ AK4 jets ( $p_T > 40,  \eta  < 2.4$ ) | $3.31 \times 10^1$ | $8.34 \times 10^1$ | $1.82 \times 10^1$    | $4.15 \times 10^1$    | $10.00 \times 10^1$ | $10.00 \times 10^1$ | $10.00 \times 10^1$ |
| $\geq 2$ AK4 jets (tight lepton veto)          | $5.98 \times 10^1$ | $9.89 \times 10^1$ | $9.28 \times 10^1$    | $9.76 \times 10^1$    | $9.97 \times 10^1$  | $9.98 \times 10^1$  | $9.98 \times 10^1$  |
| 2 muons ( $p_T > 53,  \eta  < 2.4$ )           | $1.75 \times 10^0$ | $2.69 \times 10^0$ | $4.82 \times 10^{-1}$ | $5.41 \times 10^{-1}$ | $3.90 \times 10^1$  | $3.58 \times 10^1$  | $3.23 \times 10^1$  |
| 2 muons (tight ID)                             | $8.74 \times 10^1$ | $6.32 \times 10^1$ | $1.03 \times 10^0$    | $5.10 \times 10^1$    | $8.78 \times 10^1$  | $8.03 \times 10^1$  | $7.41 \times 10^1$  |
| Muon trigger                                   | $9.36 \times 10^1$ | $9.86 \times 10^1$ | $9.19 \times 10^1$    | $9.83 \times 10^1$    | $9.96 \times 10^1$  | $9.95 \times 10^1$  | $9.86 \times 10^1$  |
| $\Delta R > 0.4$ (all pairs)                   | $1.47 \times 10^1$ | $4.58 \times 10^1$ | $7.20 \times 10^0$    | $2.54 \times 10^1$    | $9.04 \times 10^1$  | $9.35 \times 10^1$  | $9.53 \times 10^1$  |
| Jet veto map                                   | $8.10 \times 10^1$ | $7.98 \times 10^1$ | $7.75 \times 10^1$    | $8.03 \times 10^1$    | $7.92 \times 10^1$  | $7.85 \times 10^1$  | $7.90 \times 10^1$  |
| $m(\ell\ell jj) > 800 \text{ GeV}$             | $1.93 \times 10^1$ | $1.29 \times 10^1$ | $1.94 \times 10^1$    | $1.80 \times 10^1$    | $9.97 \times 10^1$  | $1.00 \times 10^2$  | $1.00 \times 10^2$  |
| $m(\ell\ell) > 200 \text{ GeV}$                | $4.76 \times 10^0$ | $6.39 \times 10^1$ | $3.15 \times 10^1$    | $1.91 \times 10^1$    | $9.85 \times 10^1$  | $9.99 \times 10^1$  | $10.00 \times 10^1$ |
| $m(\ell\ell) > 400 \text{ GeV}$                | $3.49 \times 10^1$ | $3.26 \times 10^1$ | $2.63 \times 10^1$    | $4.93 \times 10^1$    | $9.18 \times 10^1$  | $9.74 \times 10^1$  | $9.93 \times 10^1$  |

Table 14: RunIII2024Summer24: expected event yields for the resolved, dielectron channel. Each cell shows the relative efficiency (%): the fraction of events passing the previous cut that also pass the current cut. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                      | Z+jets             | $t\bar{t}$ +tW     | Nonprompt             | Other                 | (2000,1100)         | (4000,2100)         | (6000,3100)         |
|------------------------------------------------|--------------------|--------------------|-----------------------|-----------------------|---------------------|---------------------|---------------------|
| No cuts                                        | $1.00 \times 10^2$ | $1.00 \times 10^2$ | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$  | $1.00 \times 10^2$  | $1.00 \times 10^2$  |
| $\geq 2$ AK4 jets ( $p_T > 40,  \eta  < 2.4$ ) | $3.31 \times 10^1$ | $8.34 \times 10^1$ | $1.82 \times 10^1$    | $4.15 \times 10^1$    | $10.00 \times 10^1$ | $10.00 \times 10^1$ | $10.00 \times 10^1$ |
| $\geq 2$ AK4 jets (tight lepton veto)          | $5.98 \times 10^1$ | $9.89 \times 10^1$ | $9.28 \times 10^1$    | $9.76 \times 10^1$    | $9.97 \times 10^1$  | $9.98 \times 10^1$  | $9.98 \times 10^1$  |
| 2 electrons ( $p_T > 53,  \eta  < 2.4$ )       | $1.14 \times 10^0$ | $2.57 \times 10^0$ | $4.72 \times 10^{-1}$ | $5.61 \times 10^{-1}$ | $3.55 \times 10^1$  | $3.60 \times 10^1$  | $3.63 \times 10^1$  |
| 2 electrons (HEEP ID)                          | $6.73 \times 10^1$ | $4.57 \times 10^1$ | $3.93 \times 10^{-1}$ | $3.43 \times 10^1$    | $6.43 \times 10^1$  | $6.40 \times 10^1$  | $6.49 \times 10^1$  |
| Electron trigger                               | $9.86 \times 10^1$ | $9.90 \times 10^1$ | $9.42 \times 10^1$    | $9.90 \times 10^1$    | $1.00 \times 10^2$  | $1.00 \times 10^2$  | $1.00 \times 10^2$  |
| $\Delta R > 0.4$ (all pairs)                   | $1.99 \times 10^1$ | $5.05 \times 10^1$ | $2.05 \times 10^1$    | $2.80 \times 10^1$    | $8.83 \times 10^1$  | $9.28 \times 10^1$  | $9.33 \times 10^1$  |
| Jet veto map                                   | $8.26 \times 10^1$ | $8.12 \times 10^1$ | $8.02 \times 10^1$    | $8.20 \times 10^1$    | $8.17 \times 10^1$  | $7.97 \times 10^1$  | $8.09 \times 10^1$  |
| $m(\ell\ell jj) > 800 \text{ GeV}$             | $1.74 \times 10^1$ | $1.15 \times 10^1$ | $2.22 \times 10^1$    | $1.71 \times 10^1$    | $9.99 \times 10^1$  | $1.00 \times 10^2$  | $1.00 \times 10^2$  |
| $m(\ell\ell) > 200 \text{ GeV}$                | $5.30 \times 10^0$ | $6.39 \times 10^1$ | $5.10 \times 10^1$    | $2.08 \times 10^1$    | $9.75 \times 10^1$  | $9.98 \times 10^1$  | $10.00 \times 10^1$ |
| $m(\ell\ell) > 400 \text{ GeV}$                | $3.37 \times 10^1$ | $3.20 \times 10^1$ | $3.89 \times 10^1$    | $4.52 \times 10^1$    | $9.12 \times 10^1$  | $9.84 \times 10^1$  | $9.95 \times 10^1$  |

Table 15: RunIII2024Summer24: expected event yields for the boosted, dimuon channel. Each cell shows the relative efficiency (%): the fraction of events passing the previous cut that also pass the current cut. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                   | Z+jets                | $t\bar{t}$ +tW     | Nonprompt             | Other                 | (2000,100)          | (4000,100)          | (6000,100)          |
|---------------------------------------------|-----------------------|--------------------|-----------------------|-----------------------|---------------------|---------------------|---------------------|
| No cuts                                     | $1.00 \times 10^2$    | $1.00 \times 10^2$ | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$  | $1.00 \times 10^2$  | $1.00 \times 10^2$  |
| Boosted tag                                 | $9.99 \times 10^1$    | $9.76 \times 10^1$ | $10.00 \times 10^1$   | $9.99 \times 10^1$    | $9.79 \times 10^1$  | $9.87 \times 10^1$  | $9.73 \times 10^1$  |
| Leading lepton is $\mu$                     | $4.81 \times 10^0$    | $1.89 \times 10^1$ | $4.13 \times 10^0$    | $3.96 \times 10^0$    | $4.56 \times 10^1$  | $4.53 \times 10^1$  | $4.33 \times 10^1$  |
| Leading tight lepton $p_T > 60 \text{ GeV}$ | $5.83 \times 10^1$    | $8.25 \times 10^1$ | $6.55 \times 10^1$    | $7.61 \times 10^1$    | $9.98 \times 10^1$  | $9.95 \times 10^1$  | $9.91 \times 10^1$  |
| $\mu$ trigger fired                         | $9.40 \times 10^1$    | $9.50 \times 10^1$ | $9.37 \times 10^1$    | $9.39 \times 10^1$    | $9.80 \times 10^1$  | $9.74 \times 10^1$  | $9.70 \times 10^1$  |
| No $60 < m(\ell\ell) < 150 \text{ GeV}$     | $9.42 \times 10^1$    | $9.85 \times 10^1$ | $10.00 \times 10^1$   | $9.81 \times 10^1$    | $10.00 \times 10^1$ | $9.99 \times 10^1$  | $9.98 \times 10^1$  |
| Merged AK8 jet with $\Delta\phi > 2.0$      | $7.06 \times 10^{-2}$ | $1.56 \times 10^0$ | $1.48 \times 10^{-1}$ | $3.08 \times 10^{-1}$ | $6.61 \times 10^1$  | $6.44 \times 10^1$  | $5.48 \times 10^1$  |
| No extra tight lepton                       | $1.70 \times 10^1$    | $2.07 \times 10^1$ | $1.56 \times 10^0$    | $5.04 \times 10^0$    | $9.20 \times 10^1$  | $9.42 \times 10^1$  | $9.13 \times 10^1$  |
| Loose SF lepton in merged jet               | $1.00 \times 10^2$    | $1.00 \times 10^2$ | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$  | $1.00 \times 10^2$  | $1.00 \times 10^2$  |
| No loose OF lepton in merged jet            | $9.97 \times 10^1$    | $9.83 \times 10^1$ | $9.91 \times 10^1$    | $9.90 \times 10^1$    | $9.98 \times 10^1$  | $9.98 \times 10^1$  | $9.99 \times 10^1$  |
| $m(\ell\ell) > 200 \text{ GeV}$             | $9.66 \times 10^1$    | $8.07 \times 10^1$ | $6.91 \times 10^1$    | $9.23 \times 10^1$    | $10.00 \times 10^1$ | $10.00 \times 10^1$ | $10.00 \times 10^1$ |
| $m(\ell J) > 800 \text{ GeV}$               | $8.51 \times 10^0$    | $1.38 \times 10^1$ | $1.22 \times 10^1$    | $1.47 \times 10^1$    | $9.94 \times 10^1$  | $9.75 \times 10^1$  | $9.36 \times 10^1$  |
| Jet veto map                                | $8.38 \times 10^1$    | $8.24 \times 10^1$ | $7.94 \times 10^1$    | $8.51 \times 10^1$    | $8.83 \times 10^1$  | $8.66 \times 10^1$  | $8.67 \times 10^1$  |

Table 16: RunIII2024Summer24: expected event yields for the boosted, dielectron channel. Each cell shows the relative efficiency (%): the fraction of events passing the previous cut that also pass the current cut. Background estimates are from simulation; signal columns show  $(m_{W_R}, m_N)$  mass hypotheses in GeV. All yields are normalized to  $109.1 \text{ fb}^{-1}$  (2024).

| Selection                                   | Z+jets                | $t\bar{t}$ +tW     | Nonprompt             | Other                 | (2000,100)          | (4000,100)         | (6000,100)          |
|---------------------------------------------|-----------------------|--------------------|-----------------------|-----------------------|---------------------|--------------------|---------------------|
| No cuts                                     | $1.00 \times 10^2$    | $1.00 \times 10^2$ | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$  | $1.00 \times 10^2$ | $1.00 \times 10^2$  |
| Boosted tag                                 | $9.99 \times 10^1$    | $9.76 \times 10^1$ | $10.00 \times 10^1$   | $9.99 \times 10^1$    | $9.79 \times 10^1$  | $9.87 \times 10^1$ | $9.73 \times 10^1$  |
| Leading lepton is $e$                       | $4.04 \times 10^0$    | $1.61 \times 10^1$ | $3.46 \times 10^0$    | $3.36 \times 10^0$    | $4.19 \times 10^1$  | $4.00 \times 10^1$ | $3.90 \times 10^1$  |
| Leading tight lepton $p_T > 60 \text{ GeV}$ | $5.87 \times 10^1$    | $8.27 \times 10^1$ | $6.58 \times 10^1$    | $7.64 \times 10^1$    | $9.97 \times 10^1$  | $9.97 \times 10^1$ | $9.87 \times 10^1$  |
| $e$ trigger fired                           | $9.23 \times 10^1$    | $9.20 \times 10^1$ | $8.94 \times 10^1$    | $9.09 \times 10^1$    | $9.98 \times 10^1$  | $9.97 \times 10^1$ | $9.94 \times 10^1$  |
| No $60 < m(\ell\ell) < 150 \text{ GeV}$     | $9.43 \times 10^1$    | $9.85 \times 10^1$ | $10.00 \times 10^1$   | $9.80 \times 10^1$    | $10.00 \times 10^1$ | $9.99 \times 10^1$ | $9.99 \times 10^1$  |
| Merged AK8 jet with $\Delta\phi > 2.0$      | $7.90 \times 10^{-2}$ | $1.61 \times 10^0$ | $1.57 \times 10^{-1}$ | $3.37 \times 10^{-1}$ | $5.79 \times 10^1$  | $4.17 \times 10^1$ | $3.73 \times 10^1$  |
| No extra tight lepton                       | $1.81 \times 10^1$    | $2.68 \times 10^1$ | $3.72 \times 10^0$    | $8.68 \times 10^0$    | $4.66 \times 10^1$  | $2.56 \times 10^1$ | $4.02 \times 10^1$  |
| Loose SF lepton in merged jet               | $1.00 \times 10^2$    | $1.00 \times 10^2$ | $1.00 \times 10^2$    | $1.00 \times 10^2$    | $1.00 \times 10^2$  | $1.00 \times 10^2$ | $1.00 \times 10^2$  |
| No loose OF lepton in merged jet            | $1.00 \times 10^2$    | $9.80 \times 10^1$ | $9.97 \times 10^1$    | $9.95 \times 10^1$    | $9.95 \times 10^1$  | $9.94 \times 10^1$ | $9.96 \times 10^1$  |
| $m(\ell\ell) > 200 \text{ GeV}$             | $9.66 \times 10^1$    | $8.11 \times 10^1$ | $8.51 \times 10^1$    | $9.35 \times 10^1$    | $1.00 \times 10^2$  | $1.00 \times 10^2$ | $10.00 \times 10^1$ |
| $m(\ell J) > 800 \text{ GeV}$               | $9.34 \times 10^0$    | $1.09 \times 10^1$ | $1.01 \times 10^1$    | $1.97 \times 10^1$    | $9.83 \times 10^1$  | $8.91 \times 10^1$ | $8.23 \times 10^1$  |
| Jet veto map                                | $8.60 \times 10^1$    | $8.34 \times 10^1$ | $8.07 \times 10^1$    | $8.45 \times 10^1$    | $9.00 \times 10^1$  | $8.74 \times 10^1$ | $8.94 \times 10^1$  |